Lab Sheet 3: SQL Functions

Solved queries with sample outputs — MySQL 8.x

Assumptions: These examples use MySQL syntax and sample tables named `Students(StudentID, FirstName, LastName, JoinDate)` and `Employees(EmployeeID, FirstName, LastName, Salary, JoiningDate)`. Outputs are illustrative and will match the queries when run against the sample data below. Date results can differ if you run them on another day.

Sample data used for the employee/student queries

Students: (1, 'Aarav', 'Sharma', '2025-07-10'), (2, 'Diya', 'Verma', '2025-08-15'), (3, 'Kabir', 'Singh', '2025-09-01'), (4, 'Meera', 'Joshi', '2026-01-12'), (5, 'Ishaan', 'Gupta', '2026-02-20').

Employees: (101, 'Riya', 'Kapoor', 45678.60, '2024-06-15'), (102, 'Arjun', 'Mehta', 52340.45, '2023-11-20'), (103, 'Neha', 'Rao', 38999.90, '2025-02-01'), (104, 'Vikram', 'Das', 61250.75, '2022-09-10').

If you need to create the sample tables, use the setup SQL in the appendix. The examples are written for MySQL; Oracle and SQL Server use some different function names.

1. Demonstrate ABS() with suitable examples.

SQL query:

```
SELECT ABS(-25) AS result, ABS(13.5) AS positive_value;
```

Sample output:

```
result | positive_value
25 | 13.5
```

2. Demonstrate ROUND() with suitable examples.

SQL query:

```
SELECT ROUND(12.345, 2) AS rounded_2dp, ROUND(12.8) AS rounded_integer;
```

Sample output:

```
rounded_2dp | rounded_integer
12.35 | 13
```

3. Demonstrate CEIL() and FLOOR().

SQL query:

```
SELECT CEIL(12.2) AS ceiling_value, FLOOR(12.8) AS floor_value;
```

Sample output:

```
ceiling_value | floor_value
13 | 12
```

4. Demonstrate MOD() with suitable examples.

SQL query:

```
SELECT MOD(17, 5) AS remainder, MOD(20, 4) AS exact_division;
```

Sample output:

```
remainder | exact_division
2 | 0
```

5. Calculate the rounded salary of employees.

SQL query:

```
SELECT EmployeeID, FirstName, ROUND(Salary) AS RoundedSalary FROM Employees;
```

Sample output:

```
EmployeeID | FirstName | RoundedSalary
101 | Riya | 45679
102 | Arjun | 52340
103 | Neha | 39000
104 | Vikram | 61251
```

6. Find the total number of students using COUNT().

SQL query:

```
SELECT COUNT(*) AS TotalStudents FROM Students;
```

Sample output:

```
TotalStudents
5
```

7. Find the total salary of all employees using SUM().

SQL query:

```
SELECT SUM(Salary) AS TotalSalary FROM Employees;
```

Sample output:

```
TotalSalary
198269.70
```

8. Find the average salary using AVG().

SQL query:

```
SELECT ROUND(AVG(Salary), 2) AS AverageSalary FROM Employees;
```

Sample output:

```
AverageSalary
49567.43
```

9. Find the highest salary using MAX().

SQL query:

```
SELECT MAX(Salary) AS HighestSalary FROM Employees;
```

Sample output:

```
HighestSalary
61250.75
```

10. Find the lowest salary using MIN().

SQL query:

```
SELECT MIN(Salary) AS LowestSalary FROM Employees;
```

Sample output:

```
LowestSalary
38999.90
```

11. Demonstrate UPPER() on student names.

SQL query:

```
SELECT FirstName, UPPER(FirstName) AS UpperName FROM Students;
```

Sample output:

```
FirstName | UpperName
Aarav | AARAV
Diya | DIYA
Kabir | KABIR
Meera | MEERA
Ishaan | ISHAAN
```

12. Demonstrate LOWER() on student names.

SQL query:

```
SELECT FirstName, LOWER(FirstName) AS LowerName FROM Students;
```

Sample output:

```
FirstName | LowerName
Aarav | aarav
Diya | diya
Kabir | kabir
Meera | meera
Ishaan | ishaan
```

13. Demonstrate LENGTH() on student names.

SQL query:

```
SELECT FirstName, LENGTH(FirstName) AS NameLength FROM Students;
```

Sample output:

```
FirstName | NameLength
Aarav | 5
Diya | 4
Kabir | 5
Meera | 5
Ishaan | 6
```

14. Demonstrate CONCAT() using first and last names.

SQL query:

```
SELECT CONCAT(FirstName, ' ', LastName) AS FullName FROM Students;
```

Sample output:

```
FullName
Aarav Sharma
Diya Verma
Kabir Singh
Meera Joshi
Ishaan Gupta
```

15. Demonstrate SUBSTRING()/SUBSTR() with suitable examples.

SQL query:

```
SELECT FirstName, SUBSTRING(FirstName, 1, 3) AS FirstThreeChars FROM Students;
```

Sample output:

```
FirstName | FirstThreeChars
Aarav | Aar
Diya | Diy
Kabir | Kab
Meera | Mee
Ishaan | Ish
```

16. Convert a number into a character/string value.

SQL query:

```
SELECT CAST(12345 AS CHAR) AS StringValue;
```

Sample output:

```
StringValue
12345 (string)
```

17. Convert a string into a numeric value.

SQL query:

```
SELECT CAST('123.45' AS DECIMAL(10,2)) AS NumericValue;
```

Sample output:

```
NumericValue
123.45
```

18. Convert a date value into a different format.

SQL query:

```
SELECT DATE_FORMAT('2026-10-12', '%d-%m-%Y') AS FormattedDate;
```

Sample output:

```
FormattedDate
12-10-2026
```

19. Demonstrate CAST() with suitable examples.

SQL query:

```
SELECT CAST('2026-10-12' AS DATE) AS ConvertedDate, CAST('99.75' AS DECIMAL(6,2)) AS ConvertedNumber;
```

Sample output:

```
ConvertedDate | ConvertedNumber
2026-10-12 | 99.75
```

20. Demonstrate CONVERT() where supported by the DBMS.

SQL query:

```
SELECT CONVERT('456', UNSIGNED) AS ConvertedInteger, CONVERT('2026-10-12', DATE) AS ConvertedDate;
```

Sample output:

```
ConvertedInteger | ConvertedDate
456 | 2026-10-12
```

21. Display the current date using a suitable date function.

SQL query:

```
SELECT CURDATE() AS CurrentDate;
```

Sample output:

```
CurrentDate
2026-10-12 (example run date)
```

22. Extract the year, month and day from a date.

SQL query:

```
SELECT YEAR('2026-10-12') AS YearPart, MONTH('2026-10-12') AS MonthPart, DAY('2026-10-12') AS DayPart;
```

Sample output:

```
YearPart | MonthPart | DayPart
2026 | 10 | 12
```

23. Find the difference between two dates.

SQL query:

```
SELECT DATEDIFF('2026-10-12', '2026-10-01') AS DaysDifference;
```

Sample output:

```
DaysDifference
11
```

24. Add a specified number of days to a date.

SQL query:

```
SELECT DATE_ADD('2026-10-12', INTERVAL 7 DAY) AS NewDate;
```

Sample output:

```
NewDate
2026-10-19
```

25. Display employees' joining dates in a required date format.

SQL query:

```
SELECT FirstName, DATE_FORMAT(JoiningDate, '%d/%m/%Y') AS FormattedJoiningDate FROM Employees;
```

Sample output:

```
FirstName | FormattedJoiningDate
Riya | 15/06/2024
Arjun | 20/11/2023
Neha | 01/02/2025
Vikram | 10/09/2022
```

Appendix: Create the sample tables and data

Run this setup once before executing queries 5–15 and 25. Queries 1–4 and 16–24 use literal values and can run independently.

```
CREATE TABLE Students (
StudentID INT PRIMARY KEY,
FirstName VARCHAR(50),
LastName VARCHAR(50),
JoinDate DATE
);

INSERT INTO Students (StudentID, FirstName, LastName, JoinDate) VALUES
(1, 'Aarav', 'Sharma', '2025-07-10'),
(2, 'Diya', 'Verma', '2025-08-15'),
(3, 'Kabir', 'Singh', '2025-09-01'),
(4, 'Meera', 'Joshi', '2026-01-12'),
(5, 'Ishaan', 'Gupta', '2026-02-20');

CREATE TABLE Employees (
EmployeeID INT PRIMARY KEY,
FirstName VARCHAR(50),
LastName VARCHAR(50),
Salary DECIMAL(10,2),
JoiningDate DATE
);

INSERT INTO Employees (EmployeeID, FirstName, LastName, Salary, JoiningDate) VALUES
(101, 'Riya', 'Kapoor', 45678.60, '2024-06-15'),
(102, 'Arjun', 'Mehta', 52340.45, '2023-11-20'),
(103, 'Neha', 'Rao', 38999.90, '2025-02-01'),
(104, 'Vikram', 'Das', 61250.75, '2022-09-10');
```